

The Reengineering of the Common Tomato – Metallicman

 metallicman.com/laoban4site/the-engineered-fiasco-of-the-humble-tomato

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Did you know that the tomato is a fruit? It actually is, you know. It certainly does not taste like one today. Though, it really used to. Tomatoes used to be sweet and delicious.

That is, until the 1920's in the United States. What in the world was going on then? All that damned "[Progressivism](#)" and trying to change America into a utopia ruled by a [benevolent wealthy class](#). Nonsense. Oh, yes. They implemented the [Federal income tax](#), and they [banned alcohol](#) and did all kinds of things (like giving the [right to vote](#) to woman) to turn the world into "Heaven on Earth". Yup. That's why they [dragged us into a war in Europe](#) (one that our founders promised and vowed that we would never ever get tangled up into again...).

Oh, but I digress... after all, they meant well.

In part of the rush to make America a utopia, numerous programs were set in motion. One of which was reengineering the tomato. The bland, sterile, and cardboard-tasting tomato is the result of what happens when well-meaning people tries to improve something. Progressive millennials please take note.

Let me tell you all the secret story of the destruction of the humble tomato...

The Early History

From what I can gather, the tomato originated in the Americas.

They probably came from a geographic area around Peru. (Some reports place it at the extreme Southern section of South America.) As such, wild tomatoes can still be found in the Andes Mountains. We do not know if the Incans cultivated the tomato. However, we know that by the time the conquistadors came to Central and South America, there was widespread cultivation of tomatoes. I can well imagine an Incan, wearing one of those funny black hats, smunching on a tomato next to his favorite lama in the misty fog swirled mountains.

It was Spanish conquistador Hernán Cortés who first fell in love with this plant. From records, we know that he was exposed to it after he looted the Aztec city of Tenochtitlan. I can also picture him. Dressed as [Aguirre](#) and wearing the armor of that time, eating a tomato in his gloved fist; the red tomato juices running down his arm. While his armed force stands nearby with their flags and crosses. When he returned back to Europe he managed to carry some seeds back with him, along with coco, tobacco, gold, slaves...



Aguirre was the king of his domain. His domain included gold, cocaine, slaves, monkeys and tomatoes. Aguirre was the king of his domain. (Image [Source](#))

The seeds grew rapidly in the warm Mediterranean climate. For various reasons, the Spanish government started encouraging its production. As such, production began in both Europe and the various Spanish colonies. We know that during the 1540s the tomato started being produced in Spanish fields. Therefore, it can be assumed that it was begin grown as a food.

A Dangerous Poison

While the Spanish had an obvious love for this wondrous fruit, the other European nations did not.

All throughout Europe and England, it was considered a dangerous poison. It was viewed as an unhealthy plant of which the red tomato fruit was a lethal poison. Thus, instead of treating it as a fruit, they used it as a tabletop decoration. This continued through the 17th and 18th centuries. As a decorative fruit, they cherished their appearance and beauty. To get the best-looking tomatoes, they experimented with selective breeding. As a result, there was an explosion of tomatoes of many colors and shapes.

Why did everyone in Europe consider this a dangerous poison?

There are two primary reasons. One of which is the appearance. There is a striking resemblance of tomatoes to the [deadly nightshade plant](#). This is not something that can be taken lightly. Nightshade is extremely lethal. Back then, they did not have the same kinds of medical care that we have today. If you digested deadly nightshade, you would die.

Secondly, most Europeans thought that the tomato was poisonous because many of the wealthy people who ate it got very sick afterwards. This had nothing to do with the plant. However, it had everything to do with the plate that the tomato was ate off of.

It all had to do with the way plates and flatware was made in the 1500's. Rich people in that time used flatware made of pewter. Pewter is a metal that has high-lead content. What happens is that foods that are high in acid, like tomatoes, would cause the lead to leech out of the plates. So if a tomato was on the pewter plate, the lead would leach into the tomato. This of course would result in lead poisoning and death. Because poor people, who ate off of plates made of wood, did not have that problem it was obvious what the problem was; the tomato. Today we know the truth; it was a tomato on a pewter plate.

This is the primary reason why tomatoes were only eaten by poor people until the 1800's, especially poor Italians.

Changes during the 1800s

Apparently, Jefferson grew tomatoes on his large farm. As such, all of his daughters and granddaughters used them in numerous recipes. They did what many of us do today, they pickled them for storage. Jefferson's son-in-law, Thomas Mann Randolph, mentioned that though tomatoes were hardly known ten years prior, by 1824 everyone was growing and eating them. So, maybe Jefferson was an outlier, or maybe he was a trendsetter. We don't know for sure. All that we knew is that he liked tomatoes. Hey! Anyone who likes tomatoes can't be all-bad.

The popularity of tomatoes increased during the 1800's. We don't know why. I personally think that the American Civil War had something to do with it. When the [carpetbaggers](#) headed South to take advantage of the devastation caused by the war, they brought tomatoes and tomato seeds with them. Soon, the entire South was a bloom with tomato plants. The world was made right again!

| Thank God for tomatoes!

Around the same time as the [Civil War Reconstruction period](#), there was an interesting period of immigration from Italy. It is true. During the 1800's there was a mass immigration from Europe to America. Europe includes Italy, which is where the idea for pizza comes from. Indeed, how can we not talk about pizza when we are discussing tomatoes. For what is pizza without a tomato? Why it is nothing more than just bread and cheese. Indeed, there is no pizza without tomato sauce, and pizza was invented around Naples, Italy in the late 1880's.

(According to history, the story is that pizza was created by a forgotten chef in Naples. For one reason, he must have thought that it was a good one, he wanted to celebrate the visit of Queen Margarine. This queen was the first Italian monarch since the French conquered Italy. The chef made the pizza from three ingredients that represented the colors of the brand new Italian flag. These colors were red, white, and green. The red is the tomato sauce, the white was the mozzarella cheese, and the green was the basil topping. And that my dear boys and girls, is how Pizza Margarine was born.)

As popular tomatoes were, they were difficult to grow and harvest. They were mostly grown at home for personal and private use.

"There was not in the United States at the time an acre of tomatoes from which a bushel of uniformly smooth tomatoes could be gathered,"

- Alexander W. Livingston.

Oh, we know about [Thomas Edison](#), and [Nikola Tesla](#). Both of whom brought electricity to America and the world. But, what of the other great scientists and inventors and their contribution to society? Are not the others important? You betcha they are!

The so-called "modern age" of commercially grown tomatoes started as the brainchild of [Alexander W. Livingston](#). He was an American botanist and scientist who dedicated much of his life on "improving" the tomato for commercial use. In his labs he managed to cross breed different types of tomatoes. He used the selective breeding of various types for his own nefarious ends. For he wanted to make the tomato an easily farmed plant. As such he was quite successful. It was he who developed the first commercial viable tomatoes. His 1870s breed called "Paragon" became an instant success. With this, followed a large tomato industry that eventually made the tomato one of the most popular fruits in America.

Joseph Campbell of [Campbell's soup](#) came out with condensed tomato soup in 1897. With pizza and tomato soup firmly in place on Americans dinner table, the humble tomato became a solid fixture of the American diet. Everyone was eating tomatoes.

By the time the turn of the century came about (the year 1900), tomatoes were as American as [Shoe shine boys](#), [Blackface](#), [playing the piano in the home living room](#), [courting young women](#) in the family parlor (or on the front porch while they sang to them), and a [highway speed limit](#) that was established at eight miles an hour.

Ah, you can't have progress without limits, don't you know?

Enter the Progressive Utopists

Starting in the 1870s, tomato breeders have been using traditional, non-GMO breeding techniques to develop the best tomatoes. They wanted tomatoes that would look ripe (nice and red), without any green portions, blotches or soft spots on the skin. They also wanted a thick skin that would not break during picking and transport. At that time, most tomatoes were grown on small farms or at home. Tomatoes needed to be harvested often, and carefully. They fell off the vine rather easily when ripe and tended to break easily because they had a thin skin. This needed to change.

Around 1920 or so, numerous teams (often located in universities or colleges) began to implement programs to "improve" the tomato. Of course, their idea of improvement was not to make it taste better. Rather, instead to make it easier to grow, ship and sell. As such, the implementation of strong reengineering programs for the tomato began in earnest. As the tomato "improved", so did the ability to harvest it commercially.

Initially, large-scale production of tomatoes was mostly limited to the East coast of the United States. This continued, and production spread into California after World War 2. The seasonal workers in California were able to pick the tomatoes, but it would not last for long. By the early 1960's, there was a shortage of skilled tomato pickers. As a result there became a need to utilize mechanical pickers. These machines could pick the tomatoes off the vine quicker but they had some problems. Most notably, they couldn't select between green and ripe tomatoes, and they also tended to damage the tomatoes.

In those days, the machines would rip and tear the tomatoes. It would crush them. It would bruise the skin. The machines would get clogged up with the juices of damaged tomatoes, and time needed to be devoted to repairing the machinery.

Selective breeding all through the 1960's was devoted to making the tomato easy to harvest and to ship.

Tomatoes became beautiful. They became a uniform red. No green skin or splotches anywhere. They became round, instead of plump. They became firm instead of juicy. They

became thick skinned instead of thin skinned. They became perfect for harvesting and shipping.



A tomato is nothing if it is not used for smunching. I would suggest that a nice club sandwich (as pictured) be a suggested venue to eat the humble tomato. Yum!

However, all this came at a cost. They became tasteless, boring, bland and tough. They lost much of their sweetness. They lost their aromatic smell, and they lost their taste.

This was not lost on the producers who used tomatoes in sauces, stews, and soups. To compensate for the degradation of the taste in tomatoes, the industry started to add sugars and corn syrup. Thus, by the end of the 1970's tomatoes and tomato products were almost completely reengineered to be mass-produced and shipped in mass. All that remained were a few spots where "radical" holdouts treasured their own unique tomato plants. These were called "heirloom tomatoes".

As a result the tomato today is a bland, red orb. It is no longer the juicy delicious tasteful plant that it once was.

How Tomatoes became Bland

Luckily for us, other people have noticed that tomatoes have started to taste terrible. For me, of course, I have stopped buying them in the grocery store. When I buy tomatoes, I go for the smaller tomatoes, and whatever local "heirloom" tomatoes that might be available.

Other than that, I grow them next to the bamboo in my yard. Together, we have formed a silent group of tomato affectionados that demand tomatoes that have flavor.

Through study and evaluation, we have discovered why tomatoes taste so bland.

There are numerous reasons. Firstly [1], tomatoes were bred to be easy to ship. Secondly [2], tomatoes were bred to ripen evenly. Thirdly [3] they are not permitted to ripen on the vine. Finally, fourthly [4] they are bred to be plump and full of water, not full of tasty volatiles.

Easy to Ship

It turns out the problem with tomatoes today is that the genes that give tomato taste has been bred out of the plants. Over the years, the tomatoes have been bred to behave more like a shipping box, or a cardboard box. Of course this was done for shipping concerns. In the past, tomatoes had a hard time surviving the road to the market. They needed to be bred to act like a shipping box, or cardboard box. Which pretty much explains why they taste that way as well. Duh!



Tomatoes were bred to ship like a cardboard box. They were designed to be sturdy and have a maximum shelf life, as well as a long display time. Indeed, the ideal tomato, from the point of view of a store, is a cardboard box.

Researchers into this problem have discovered a “genetic switch” within the tomato. This switch is responsible for much of the sugar production within the tomato. In the breeding efforts to make the tomato easy to transport to the grocery store, the growers have turned off that switch. All the efforts to breed the tomato for hardness destroyed the ability for the tomato to taste good. It bred out the sugar. This can make chewing on the tomato like eating a piece of rubber. The mutation to make the tomato act like a shipping box inadvertently turned off the sugar production switch, making the tomato bland.

Researchers discovered that a protein called GLK helps to create the sugar within the tomato. It also regulates photosynthesis in the plants, which also contribute to the production of sugar. This protein can be found near the stem and also tends to delay the reddening of the tomato skin. So all those nice red tomatoes in the supermarket contain

either less or no GLK proteins, meaning the tomatoes taste bland. Yes. So if you see a nice evenly red tomato, smell it. If it has no smell, it will also have no taste.

Now comes the frightening part.

It's not just in the United States. It's all over the world! Indeed, researchers tested over 25 commercial tomato varieties from all over the world. Every single one of them contained the genetic switch mutation. None of them had any decent quantities of the GLK protein.

"The mutation they describe in their paper is in literally 100 percent of modern breeds sold in grocery stores today,"

-Harry Klee, a molecular geneticist at the University of Florida, Gainesville

Even Ripening

Just like the genetic switch to make tomatoes easy to ship, a genetic switch was found to make them ripe evenly.

As stated previously, commercial tomatoes are selected for uniform ripening. It was discovered early on that shoppers would select the ripest and reddest tomatoes in the grocery store. People bought red tomatoes. They shied away from green tomatoes, and tomatoes with imperfections. People came to associate red tomatoes with ripeness and large quantities of sugar. This is what happened, even though that is not what was actually happening.

Today, unripe (commercial) tomatoes are a light green color that turns into a uniform red when ripe. This is how it is with the breeds today. However, this was not always that way.

Heirloom tomatoes grew differently. When they ripen they do not do so evenly. They often have a dark green coloration near the stem even when the fruit is fully ripe. This is the secret to flavor. This top area of the tomato is known as the shoulders. It is what makes the tomatoes tasty.



Green shoulders on tomatoes are what makes them taste good and sweet. It is too bad that this feature has been bred out of commercial tomatoes.

The big problem, and the reason why it was genetically bred out, is that green shoulders make it harder to tell when the fruit is ripe and ready to pick.

Shoppers and grocery stores were often unhappy with green shoulders. That is, until 1937. In 1937 a “miracle tomato” appeared. This tomato, called “All-red” had the curious ability to ripen evenly. It was bred at the North Dakota State University Agricultural Experimental Station. The “All-red” tomatoes ripened to a bright uniform red with no green shoulders. This trait was called the ‘uniform ripening’ trait. It revolutionized the world of tomatoes and was immediately incorporated into all of the commercial tomatoes worldwide.

Yah. But like any horror movie, there are tradeoffs that you need to confront.

The so-called ‘uniform ripening’ trait avoided green shoulders by turning off a very important and key tomato gene. This gene, known as *SlGLK2*, increases chloroplast production in the tomato plant. Chloroplasts make energy through photosynthesis. They also turn tomato fruit a darker green. They ripen more evenly, which makes them more visually pleasing to consumers, and it’s easier to tell when they’re mature. This trait is governed by a cluster of genes called the “uniform ripening locus” or simply “u”. But...

But...

These chloroplasts tend to make the important stuff. They manufacture sweet sugars and yummy carotenoids. In fact, the non-commercial tomatoes; the heirloom tomatoes receive 25% of their sugars from the chloroplasts in the fruits and 75% from leaves. So surprise, surprise! Those green shoulders on tomatoes are actually a good thing. They are

a visual indicator that the tomato has more carbohydrates and more flavors. By breeding out the green, growers were breeding out the flavor.

Ignore the Vine

Today's commercial tomatoes ignore the vine. The tomatoes are not permitted to ripen on the vine, which as everyone knows, is the best way to enhance flavor in fruits and vegetables.

Instead, they are picked green and stored in the dark. In order to give them that nice uniform red color, the unripened green tomatoes are gassed with ethylene. This is a nice trick that modern farmers use. It is a way of painting the tomatoes a red color without using red paint. The ethylene gas reacts with the tomato skin. When gassed, the tomatoes develop a red coloration. Which is, of course, the appearance of ripening.

Store bought tomatoes are not ripe. They just look like it. Real tasty tomatoes are vine ripened. Vine ripened tomatoes develop more sugars and thus more flavor.

Finally, all (if not most) commercial tomatoes are usually refrigerated. Refrigeration damages both fruit flavor as well as texture.

Full of Water instead of Volatiles

Aside from the tasty sugars, modern tomatoes were bred to be full of water. The volatiles were bred out of them. The key to a good tasting tomato is the volatile compound. This is a chemical that easily becomes a gas and therefore can be smelled. It is the volatiles, which are sorely lacking in most mass-bred tomatoes. They are a primary source by which a tomato acquires its sweet taste.

Now, the stores that sell the tomatoes, and the farmers that grow the tomatoes do not care at all about the taste. They want the biggest and heaviest tomatoes possible. That is because the heavier the tomato, the more money that they can sell it for.

Once a truck is loaded with tomatoes at the farm, the buyer of the tomatoes weighs the truck with and without the tomatoes. The difference is the weight of the tomatoes. The heavier the tomato, the more money the farmer makes.

This is a case of bigger is better. So the breeders turned on a combination of genetic switches that turned off volatiles, and bred the tomatoes to be water containers.

Volatiles were bred out of most store-bought tomatoes. Instead of volatiles, water has become the dominant component. What this means is that not only do mass-produced tomatoes don't taste very good, but they don't have as much nutritional value either.

There are six major volatile compounds that add the important flavor to a tomato. This was identified by Harry Klee, a biologist at the University of Florida. In research, he and his colleagues identified six volatile compounds that work together to impart better taste in a tomato. What is most interesting is that they do not particularly create sugars. Instead, these compounds add a sweet taste to the tomato without adding sugar.

Klee, experimented with 100 tomato varieties with 13 panels of 100 people. Each one rated each tomato according to taste and sweetness. The researchers used a technique known as gas chromatography to vaporize the tomato samples and sort out their molecules. This resulted in a list of chemicals and their concentrations within each sample. It was through this study that they found out that chemical compounds known as volatiles are very important. Volatiles are released as we chew and trigger a response in the olfactory system, contributing to overall taste sensation. By comparing the consumer panels' tomato preferences to their chemical profiles, the team came up with a list of 13 chemical compounds strongly linked to likability.

Surprisingly, much of that taste came from the volatile compounds and not from sugar. Klee found that the volatiles, working through sensors in the nose, convinced the brain that a certain tomato was sweeter than another. Thus, when a tomato has both volatiles and natural sugars, the taste in a tomato really pops out.

Unfortunately, volatiles have been bred out of all the commercial tomato varieties.

Today

Today all of the commercial tomatoes no longer make sugars in the tomato meat. They get them from the leaves. They are unable to make their own sugar because the SIGLK2 has been disabled. They have no taste because they have no volatiles. They produce less carotenoids than heirloom tomatoes. This means that they have no interesting flavors (sweet or not sweet). Commercial tomatoes have less carbohydrates, less carotenoids, no volatiles, and less sugar.

But they do look better.



The ideal tomato from a vintage advertisement. Nice big, plump and red with no green shoulders, no imperfections, and no taste. ([Image source.](#))

Enter the Busybodies

Much as I hate to say it, there are a number of busybodies that want to destroy the chances of you (the reader) ever getting to eat a “real” tomato. [The EU](#), out of Brussels [wants to ban it](#). Leaving everyone in the EU stuck with eating cardboard tasting commercial GMO “improved” tomatoes. They have come up with excuses to justify this decision, but as you should know by now, there are reasons and then there are true REASONS for doing anything.

In America, certain liberal progressives are out at it again. They too want to [ban Heirloom tomatoes](#). According to [Jane Black](#), commercial tomatoes taste just as good, if not better, than the heirloom tomatoes. Of course, I disagree. I think it is because I have a greater appreciation of food than those busybodies. But, you know, it’s only my opinion. Ah, but what else would you expect from WaPo? Eh?



Modern Tomato Vs. One Grown From 150-Year-Old Seeds

Modern tomato compared to the “improved” progressive tomato

Ah, like all busybodies, they have their excuses. However, like all busybodies, they tend to keep their real reasons secret...

What Can We Do?

The first thing that I would suggest is to start ordering and planting heirloom tomatoes at home. In the summer, stop buying them at the grocery store. Every chance you get, complain to the grocer and the manager of the vegetable section how terrible the tomatoes taste and that you refuse to buy them. If enough people do this, perhaps we can one day reverse this deplorable situation.

I buy mine off [taobao](#). The link will take you straight to the tomato seed section. Remember that one USD dollar is around 6.3 yuan. So take the cost and divide by six to figure out how much the seeds cost in dollars.

I am curious what others think about this. Perhaps you too have noticed that store-bought tomatoes taste like water filled baseballs. Maybe there are people who have grown their own tomatoes and have some hints on how much better they are. Maybe there are people out there in internet-land who, like me, also love the tomato sandwich. Come on. Don't be shy.

I would also like to know about your tricks on planting seeds. We cannot buy tomato sprouts here in China, and I am just having a real dickens' of a time trying to get them to

grow. I've followed everything recommended [here](#), and [here](#), but they just sort of never make it past germination. Right now, today at the time of this writing, I have a plastic egg container filled with chicken poop with various Chinese heirloom seeds in it. I keep them moist and in the sun. It's been ten days, and still no sprouts. Why?

—UPDATE—

It turns out that seeds in China are all coated with a plastic paint. This prevents the seeds for germinating in hot and humid weather. You need to scrub the paint off prior to planting the seeds. You let the seeds soak in warm water for around ten hours, and then you rub the paint off. Then once the paint is off, you can then plant them.

—END UPDATE—

It's a tough world out there, and tomato lovers need to stick together against the onslaught of tomato haters, busybodies, and the armies of ignorant fools.

I believe, and I say, there are certain things that we need to fight for. These things are important to us, and we need to draw a line in the sand. I say that if we cannot have a tasty tomato, then what is the point of eating them! Gosh Darn It!

Links

Here are some links to buy some great heirloom tomato seeds;

- [Heirloom tomato seeds for purchase](#). How to grow heirloom tomatoes. All varieties non-GMO.
- [Baker Creek Heirloom Seeds](#) offers 200+ varieties of rare heirloom tomato seeds.
- [Burpee's heirloom tomatoes](#) are proven to have an aroma, flavor and freshness money can't buy.
- [TomatoFest is the ultimate resource for organic heirloom tomato seeds](#) with over 600 organic heirloom tomato seed varieties along with growing information.
- A [good selection](#) of heirloom seeds.
- [Annies Heirloom Seeds](#).
- [Eden Brothers](#) Heirloom Tomato seeds.
- [Gary Isben's](#) Tomato Seeds.
- [Tomato Growers Supply](#).

Conclusion

One of my all-time favorite things to do, in the summer, is to enjoy a tomato sandwich. This is a very simple affair. It is just a big slice of a big juicy "[Big Boy](#)," or "[Better Boy](#)" tomato between two slices of plain white sandwich bread and gobs of [Miracle Whip](#) salad dressing.

Unfortunately, tomatoes have been getting less tasty over time. Initially I didn't notice it so much. But when you taste a "real" tomato and then experience a "commercial tomato" the difference is astounding. Look, I don't just mean "Hey, that's a pretty tasty tomato.", I mean "WTF! WTF! Holey Heck! What is the Friggin' heck going on with this tomato!" It is that noticeable.



Talk about Heaven on Earth! How about properly cooked bacon on rye bread, with fresh heirloom tomatoes, with salt and pepper and Miracle Whip . Yum! ([Image Source.](#))

Those that have written about this phenomenon tend to down play the significance. They shouldn't. It's a serious issue.

While I might joke about the millennials who want a utopia where unicorns prance about on fields of clover in the bright sunny light, I too wish for a perfect life. However, my idea is a much simpler one. I dream for a nice fresh heirloom tomato sandwich on Miracle Whip slathered bread and an frosty cold bottle of a good golden beer. I can only pray that this becomes a commonplace reality.

There is Hope

From an article from Sci-News;

Tomatoes are one of the most eaten vegetables — although they actually are fruit botanically — with a worldwide annual production of 182 million tons, worth more than \$60 billion.

U.S. tomato consumption per capita was 9.2 kg (20.3 pounds) for fresh tomatoes in 2017 plus an additional 33.2 kg (73.3 pounds) of processed tomatoes eaten per person. They are the second most consumed vegetable in the United States after potatoes.

While cultivated tomatoes have a wide range of physical and metabolic variation, there have been several severe bottlenecks during its domestication and breeding. This means today's tomatoes have a narrow genetic base.

The tomato pan-genome helps identify what additional genes beyond the reference might be available for crop breeding and improvement. It includes all of the genes from 725 different cultivated and closely related wild tomatoes, which revealed 4,873 genes that were absent from the [original reference genome](#).

"The pan-genome essentially provides a reservoir of additional genes not present in the reference genome. Breeders can explore the pan-genome for genes of interest, and potentially select for them as they do further breeding to improve their tomatoes," said co-lead author Dr. Zhangjun Fei, a researcher with Boyce Thompson Institute for Plant Research and the U.S. Department of Agriculture-Agricultural Research Service.

"One of the most important discoveries from constructing this pan-genome is a rare form of a gene labeled [TomLoxC](#), which mostly differs in the version of its DNA gene promoter," added co-lead author Dr. James Giovannoni, a molecular biologist with Boyce Thompson Institute for Plant Research and the U.S. Department of Agriculture-Agricultural Research Service.

"The gene influences fruit flavor by catalyzing the biosynthesis of a number of lipid-involved volatiles — compounds that evaporate easily and contribute to aroma."

In addition, the researchers found a new role of TomLoxC — it facilitates production of a group of [apocarotenoids](#) that work as signaling molecules influencing a variety of responses in plants including environmental stresses. The compounds also have a variety of floral and fruity odors that are important in tomato taste.

The rare version of TomLoxC was found in only 2% of older or heirloom cultivated large tomato varieties, although the version was present in 91% of currant-sized wild tomatoes, primarily [Solanum pimpinellifolium](#), the wild predecessor of the cultivated tomato. It is becoming more common in newer varieties.

"It appears that there may have been strong selection pressure against or at least no selection for the presence of this version of TomLoxC early in the domestication of tomatoes," Dr. Giovannoni said.

"The increase in prevalence of this form in modern tomatoes likely reflects breeders' renewed interest in improved flavor."

The [results](#) are published in the journal *Nature Genetics*.

Lei Gao *et al.* The tomato pan-genome uncovers new genes and a rare allele regulating fruit flavor. *Nature Genetics*, published online May 13, 2019; doi: 10.1038/s41588-019-0410-2

Take Aways

- Tomatoes are a fruit.
- Tomatoes taste like a vegetable because the sweet taste was bred out of them.
- The reason that tomatoes taste this way is due to greed.
- Most people, and science agrees, that heirloom tomatoes are tastier than commercial tomatoes.
- Liberal progressives in the EU and America want to ban tasty tomatoes.
- I like fresh tomato sandwiches and icy cold beer.

RFH

How about a Request For Help? I tire of busybodies and statists who poke fun at the ideas and theories of others. They offer no constructive dialog. Rather they just make fun, ridicule, and then scurry under a rock.

I use this forum as a way to disseminate some of the things that I learned though my years of life. I like to talk about things as I have experienced them, and I am always willing to listen to what others have to say. (Though when it comes to tomatoes, I do have some very strong opinions.)

So, if you, the reader, were so interested, I would welcome your stories. Tell me how you came to love tomatoes, and how to grow them. Tell us your secrets on getting the sprouts started, because that is one thing that I am having a very difficult time doing. Tell us all how you use them, and why you think that it is just “OK” to ship and sell cardboard tasting tomatoes in stores. Share with us your theories as to why the EU is trying to ban any seeds that are not being used by the top food-producing mega-corporations. Share with us, because we want to know.

This is my callout, to you the reader, to assist all of us in solving these mysteries. After all, this is a far better use of the internet than for looking at Justin Bieber videos.

FAQ

Q: What is a “heirloom tomato”?

A: A heirloom tomato is a tomato variety that was NOT bred for mass consumer sales and

distribution. As such, it was bred for taste, appearance, and smell. Because heirloom tomatoes are difficult to pick and ship for the mass market, they are seldom found in grocery stores. They can (for the most part) be obtained by growing them yourself by planting seeds.

Q: What is the history of the tomato?

A: Tomatoes originated in South America near Peru. They have been bred over the years as a consumer product. As such, they are now treated as a vegetable instead of as a fruit. This is because the sweetness and volatiles have been bred out of them. They are used in the most important food dishes in America; the pizza and hamburger.

Q: What is a tomato festival?

A: A tomato festival is a place where people display and sell their own (often non-commercial) tomatoes. These are various home-developed heirloom tomatoes. It is an excellent place to sample different types of tomatoes, chat with other tomato lovers, and buy and trade tomato seeds.

Q: Why do you like beer?

A: Icy cold beer tastes great after a long hard day of work and labor. It relaxes you. It lets your tension go away, and it tastes really, really good with fresh heirloom tomatoes. (Oh, so does wine, but you drink it differently. Hillary Clinton is a big Chardonnay fan. I can't fault that, she does have good taste in wines. Though I am a dry red wine fan, myself.)

Q: Can I drink wine with a tomato sandwich?

A: Let me say this clearly. Tomatoes taste good anytime, anywhere, and prepared in any way. Wine goes great with just about everything. Hillary Clinton is a big Chardonnay fan. I can't fault that, she does have good taste in wines. Though I am a dry red wine fan, myself.

Q: How do the Chinese eat tomatoes?

A: The Chinese people love tomatoes just like Americans do. They use the tomatoes for all sorts of dishes, but the most important one is the "[tomato with egg](#)" dish. This is an AMAZING dish that every single American who has visited China has fallen in love with. It is not what you think; just scramble eggs with tomatoes. No. You prepare the tomatoes with some special spices and then mix in eggs. It is glorious!

Q: What is the best use for a tomato?

A: I think that the best use for a tomato is to eat it. I really don't think that it can compete against the rose as a decoration. When eating it, I would suggest that it be picked off the vine fresh. I do prefer vine ripened tomatoes. A good tomato tastes great just as is. I often add a pinch of salt and pepper for flavor, but that is just me. If you need to add sugar, then you know that the tomatoes you are eating are a commercial variety.

This article was posted on free Republic on 5AUG18. You can read the comments [HERE](#).

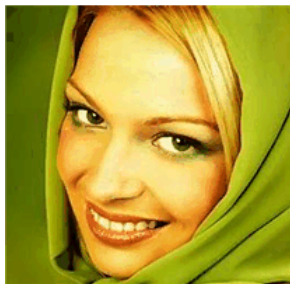
Posts Regarding Life and Contentment

Here are some other similar posts on this venue. If you enjoyed this post, you might like these posts as well. These posts tend to discuss growing up in America. Often, I like to compare my life in America with the society within communist China. As there are some really stark differences between the two.



Why Can't America have High-Speed Bullet Trains?

Well? Why not? Just about every other nation of any significance has them. But, somehow, America just can't afford them. America spent over 77,000,000,000 on a high speed rail line in California, and only got 15 miles of track laid down, no stations, and some multi-million dollar fiberglass and cardboard mockups.



The Most Important Things That a Man Looks for in a Woman.

Here, we look at the things that men search for in a woman. These things are considered, by men, to be absolutely essential in a mate. We discuss ten attributes and go into detail as to why they are important.



Adventures Exploring the Cemeteries of Indiana

Cemeteries can tell us a lot about ourselves, our history, and our future. This is most especially true in regards to smaller and remote cemeteries located in the counties that surround us. This is my story about the impressions that my forays into these curiosities made upon me.



What it was like being a kid in the early 1970's

This is my narrative of what it was like growing up a a young teenager in 1971. This took place in the hills of Pennsylvania outside of Pittsburgh. I was 13 years old. I went to school, rode my bicycle around everywhere. I fished with my father, and watched television a lot.



The Fiasco of the Reengineering of the Common Tomato

This is the sad, sad story about how the wonderful and tasty tomato was reengineered to be “better”. It was made better to ship. Better to sell. Better to store. However, there was one thing missing. No one paid any attention to taste, which is why it tastes like a cardboard shipping box today.



A Movie Tribute to the 1980's When Ronald Reagan Ran America.

Here we argue that movies reflect society when they were made. Today we see popular movies being remade as all female versions due to the progressive realities today. Likewise, the movies of the 1980s reflected the time when they were made as well.



Mad Scientist Explorations

Here we discuss what it was like growing up as a boy in the 1960's and the 1970's and the people, things, and adventures that inspired us. As such, we explore how the independence of growing up enabled us to cope through life when we got older.



The Gorilla Cage in the Basement

When I was a small boy, I used to play in the abandoned buildings nearby. Here is a story about one of the objects that lay hidden inside one of those buildings and the possible mysteries that it held. As a child we thought it was a gorilla cage, now as an adult, I am not so sure...



The Two Fundamental Types of Families and how they work

One of the reasons why there is apparently so much strife and upset in the United States today is because the federal government has permitted the destruction of the fundamental family unit. This destruction has resulted in the various conditions that are prevalent in America today.



The Importance of a Family Meal Together

There is nothing more important and satisfying than spending time together with friends and family and eating a meal together. It is a time of friendship, familial bonding and relaxation. Here we discuss what I have learned from my childhood and how I apply it today in my life.



The Pleasures of Fresh Baked Bread with Butter

One of the things that I have come to miss, since I moved to China, is fresh hard crusty rolls with full salted butter. The Chinese are not really fans of crusty bread. They like sweets and sponge cakes. Here I describe something that is close to me; not appreciating what you have right in front of you.



What Work was Like in the 1960's and 1970's.

When people go to work today, they find that there are all kinds of rules and regulations. It is for diversity, safety, inclusion, and improvement in a new progressive world. Well, here is what work was like before it became a sterile progressive testing ground for the New World Order.



Learning during my 1970s High School years

Here is a pretty long post on what it was like to go to elementary school in the 1960's, and then attend high school in the 1970's. I graduated in 1977, and that my friend is exactly the target date for the famous movie "Dazed and Confused". Here is the real deal what it was like.



Some True Stories of Cat Heaven

Being in MAJestic taught me many things. One of which was an intimate understanding of Heaven and other "spiritual" things relative to quanta entanglements. Here I discuss what it is like to have a beloved animal (cat) dies and what actually happens to them and why.



Life Lessons from Working within the Corporate Dream – 1

After I was trained by MAJestic, I was left to fend for myself in Corporate America. It was a hellish existence. As such, I learned many lessons concerning behavior and the ultimate impact that it holds on our families, our friends, our loved ones, and our very lives. Here are my commentaries.



Life Lessons from Working within the Corporate Dream – 2

This is part two of the two part section. There is quite a lot of information to impart. Stories are told, and experiences relayed. Honestly working in corporate America in the 1980's, through the new century really sucked. It was harsh on everyone except the owners of the companies.



How to Build Up Your Life from Nothing.

This is for all those young men who are just leaving school and trying hard to build up their life. You can do it. Here are my thoughts and advice for a young man who is encountering that most difficult of challenges; carving your life out in the wilds of the world.



Allow your Children to Play and Grow (Part 1)

I am horrified, just horrified, that parents do not allow their children to play, roam and explore on their own. It has created a society of pampered fearful children. Here I discuss this in terms of what I know. This is very non-PC and might offend the more liberal readers out there.



Allow your Children to Play and Grow (Part 2)

This is the second part of my rant about permitting children to play, explore and have fun. Everyone, the odds that some stranger will abduct your child is very small. Most people who do this are family friends and neighbors. Stop being so fearful. It harms the development of your child.



Ode to Diabolical Cretin John McCain

Well, that evil son-of-a-bitch is finally dead. If his grave wasn't so far away, I go there and dance on top of it. This guy has spent his life backstabbing people, hurting people, concocting wars, and doing everything in his power to disrupt the lives of people. Good fucking riddance.



The Song "Baby Got Back" Translated into Latin

Here is a fun post. The song "Baby Got Back" is translated into Latin and then back again. I have to tell you that it is indeed hilarious. *Rebecca, ecce! tantae clunes isti sunt!* (Rebecca, behold! Such large buttocks she has!) Anyways, it's just all fun and games. Enjoy.



Every Man Should Have a Roll-Top Desk

It's true, and I discuss why. A roll-top desk is made for the person who accumulates the clutter of the road as they forge their way through life. The desk becomes a symbol and a place for a man's privacy. It is a place where he has control over his life, and a place to store his treasures.



Why Women Need their own Personal Makeup Vanity Space

I am a strong believer that everyone needs privacy and a personal space to call their own. In this post I discuss the need for a woman to have a vanity and why it is important for a woman to spend time using it. Everyone, regardless of their situation, needs a place that is all theirs and theirs alone.



Democrat Busybodies and the Destruction of Freedom

Wouldn't life be wonderful if you could just be left alone and live your life in peace? I think so. The founders of the United States thought so, however there is an army of busybodies who disagree. They have this vision of utopia where everyone listens to their commands.



Adventures of a 70's Kid in an Army And Navy Store

Here we look at what adventures an Army and Navy store could hold for the inquisitive child. We talk about the history and the inherent adventures that lay hidden in dusty boxes, and dark corners. We narrate the exploration through the eyes of nostalgia. It's a pleasant excursion.



Comparison between American and Chinese Playgrounds

The differences between American and Chinese playgrounds couldn't be more telling. It is a tale of how things are handled differently and the roles that parents take. In China the role of the parent is to equip the child for life. While in America it is to protect the child from life.



Excuses We Use That Keep Us Enslaved

Here we look at what an American *actually* is. For the argument made here is that citizenship in the United States is not what qualifies a person to be an "American". Rather it is the way a person thinks, the way they act, and their general philosophy in life. For not everyone is an American.

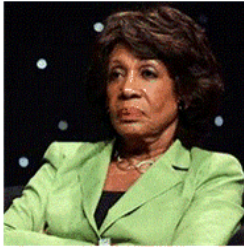
Posts about the Changes in America

America is going through a period of change. Change is good... that is, after it occurs. Often however, there are large periods of discomfort as the period of adjustment takes place. Here are some posts that discuss this issue.



Moma's Chicken Soup - A Parable About Modern America

This is a parable about modern America. It takes place around a global staple; the humble chicken soup. It compares American political news to that of making chicken soup, and utilizes many of the same catchphrases and memes in the process. It is a tongue in cheek look at America today.



What the Progressive Liberals have in store for Conservatives

The progressive socialists, who now go under the name of "Liberals", have plans on the consolidation of their power and control. This is what their ultimate plans are. It does not bode well for traditional American values, Rights, the Constitution, or history. It's going to be a dangerous time.



What the Progressive Socialist Liberals have in store for Conservatives (part 2)

When the American second civil war occurs (CWII), it will be an ideological genocide. It will follow the identical patterns that have happened in all of the genocides of the twentieth century. How do we know? Because the identical pre-genocide preparations have already been put in place.



What the Progressive Socialist Liberals have in store for Conservatives (part 3)

Everything is currently in place for a full-scale genocide against "the privileged white", the "rural deplorables", and the "crackers who are racist and love slavery". You know who they are targeting. It is cultivated, and intentional to drum up hatred and zeal to "transform America".



What the Progressive Socialist Liberals have in store for Conservatives (part 4)

The Rwandan genocide was initiated, and put into motion, by the rich urban elite inside the cities. They used the Civil War as an opportunity to "cleanse" the nation of thought contrary to their ideology.



What the Progressive Socialist Liberals have in store for Conservatives (part 5)

When SHTF, and CWII occurs scenes like this will be common place. Young armed youth will travel forth from the cities in new trucks right off the sales lot, and they will attack the people in the smaller cities and rural areas. They might even have technicals installed on brand new trucks.



What the Progressive Socialist Liberals have in store for Conservatives (part 6)

We conclude this study on history and what could very well happen in the United States, by looking at what happens to girls, and women during these contentious times of societal upset. We review the role of the Bill of Rights in protection.



Calexit and the American Civil War

This is a pretty detailed look at the causes of the American civil war, and why it wasn't about slavery. We look at the rewriting of history by the internet, and the efforts that President Obama took to keep California intact so that the Democrats could maintain political power and control.



The Warning Signs that Government Collapse is Imminent

All nations go through cycles. They do so, and when they mature they evolve into other governmental forms. The United States is long overdue for a change; a reset. Now, while this will occur, it is unknown how severe it would be. Here are my thoughts on this issue and some warnings.



The r/K Selection Theory applied to American Politics

The r/K theory is a pretty well-known theory on animal behavior. It simply states that animals behave as they need to based on the scarcity of resources. Here, we argue that it can be applied to why humans follow different political philosophies. Especially in American politics.



The Line in the Sand – Now What?

It's the same story. Just vote for us, and we will correct everything. Yet, nothing ever happens. You know, boys and girls, this old "song and dance" can only be maintained for so long. Eventually it will fracture, and people will no longer rely on elections to institute change...



The importance of having a second passport.

So President Trump is building a wall to protect Americans. Hum. Where have I heard that before? Oh yes, in Germany right when they were making the wall that separated Berlin. Walls serve two purposes; they keep people out, but they also keep people in. Be careful what you wish for.



How to Build a Gallows and a Hangman's Noose

The United States is entering a period of strife. There will be violence. There will be discord and discomfort. It will be an ugly time. There will be winners and losers. It will be painful for the losers, and the victor will need build a gallows. This is how to do so and why it is needed.



Some things that need to be changed for America to be the "Land of the Free" again.

There are many things that a person can do to improve themselves. But, what about a nation? What about the United States? Here is a handful of suggestions that I would like to propose to help America become whole again.

More Posts about Life

I have broken apart some other posts. They can best be classified about ones actions as they contribute to happiness and life. They are a little different, in subtle ways.



On Being an Older Gent

Here are my thoughts on growing old as a man. The truth is that the life that I expected to have as I was growing up never materialized. Instead, something else manifested. My life today is nothing compared to what I thought it would be. For my illusions were created by American media.



Things that I wish that I knew when I was 25 years old

Here is the advice that I would give myself when I turned 25 years of age. It is the advice I would give my younger self after attending the school of "hard knocks" for around four decades or so. If only I knew then, what I know now. Would my life be any different? Let's look at this curiosity.



Be Who You Are

All our lives we are told who to be, what to do, what career to take. Our religion, our work, our schools, our government, and our loved ones all tell us what we need to do. Here, I tell you the opposite. Stop allowing others to tell you what to do. You know what you need inside of you.



Why an American man should leave America and Travel

All people should get up off their duff and go forth and Travel. However, this is most especially true for American men. The United States has become a crucible of PC progressive realities that attack and depreciate American men. Go out, young man, see the world as it really is!



The Amazing Bremelanotide PT-141

Forget Viagra, and all those other ED medications. This is the real deal. This little baby causes humans to *go into heat*. It interacts with the brain and sets up feelings of intense arousal on both men and women. It turns people into 16 year olds. As such it is banned in the USA by the FDA.



Bronco Billy and the 25th Law of Power

Bronco Billy is a fictional movie starring Clint Eastwood. The movie depicts the life of a man who decides to change his life and go ahead and do what he has always dreamed of doing. He goes out and becomes his dream. There are lessons here for all of us, we only need listen.



How they get away with it

Ten years ago, Bankers committed fraud in Wall Street that just about devastated the nation. Earlier, we had the S&L fiasco, the nightmare with penny stocks, and now the grand wholesale treason with the selling of Top Secrets to a foreign nation. This is how they get away with it.



How Rocket Scientists Build Paper Airplanes

This is a tale of what me and my classmates used to do when we were young Aerospace Engineering students at the university. We would design paper airplanes and shoot them down with bottle rockets. Here is just a fun little exercise down “memory lane”. Enjoy.



What is Snopes.com and can it be Trusted?

Snopes is a “fact check” website that is fully funded by the oligarchy. It is used to “shout down” any narratives that does not support the rich and powerful, and politically connected globally. Here we discuss the organization, who they are and what they represent.



Today America Taxes it's Expats without Representation.

This is a sad state of affairs. Americans who reside outside of the United States now have to pay American taxes, even though they have no representation in Congress. As I recall this was one of the reasons for the American cry for independence in 1776.



What life is like inside the ADC Prison in Arkansas

This post discusses what life is like in a hard labor prison in Southern Arkansas. We discuss hoe squads, food, the hole, commissary, dress, fashion, homosexual culture, prison gangs, murders, relationships and other aspects of life when you are sentenced to "Hard Time".



The Hazing of New Employees – A Lost Tradition

Up until the 1980's all companies would haze new employees. Today, we live in a world regulated by the government and policed by Human Resources. We forget how things used to be just a handful of decades ago. Here is my narrative when I started to work.



A Wish List of Favorite Firearms

This is a personal list of many favorite firearms that I would like to own. There are various reasons for the particular weapon of choice, and I provide my reasons. This post has no other purpose than to express my desire to own certain weapons that I cannot as an American.



The Ultimate Method to Make Your Computer Safe and Secure

The best way to make your computer secure and private is to have very strong encryption and use an obscure operating system that very few people know about. Here we look at 37 obscure computer operating systems.



Is America a Republic or is it a Democracy – What is it?

Here we take a nice good look at what the United States is today. We look at reports that are suggestive of other governmental organizations. We look at what is going on contemporaneously inside the United States, and we look at how the United States behaves globally.



The Many Things that Democrats want to Ban

This is a running list of the many things that American Democrats want to ban, have banned, or are trying to ban. You know, a ban of anything is a restriction on freedom. That is not what America was founded upon. Yet, here we are today. Everyday is a new ban, and loss of freedom.



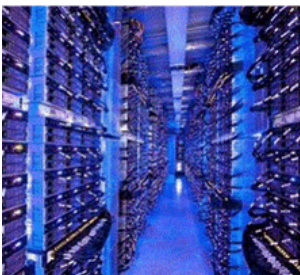
Nations that an American Man could Travel and Relocate to – Part 1

Here, I suggest places where an American man could go if he wanted to start all over again. You know, this happens. Often a man will go through a nasty divorce, or get raked over the coals by a business associate or the United States government. When it happens, it's time to start new.



Nations that an American Man could Travel and Relocate to – Part 2

Here, I suggest places where an American man could go if he wanted to start all over again. You know, this happens. Often a man will go through a nasty divorce, or get raked over the coals by a business associate or the United States government. When it happens, it's time to start new.



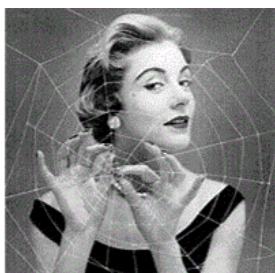
Alternative Search Engines instead of Google

Google does two things that I do not like. Firstly, they vacuum up everything you do and send the information straight to the American deep state; the NSA and DHS. Secondly, they censor your results for you according to what they think you should see. Here we look at alternatives.



The Consequences of the #MeToo Movement

While the #MeToo movement burst forth onto American culture with good intentions, it was soon coopted by those with selfish and sinister intent. The aggressive nature of the accusations, and the consequences of being accused became dangerous. Here we look at the corporate aspect.



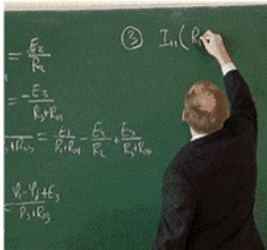
The Lies the Media Tell Us

The United States mainstream media does not report "news". Instead they fabricate events that are designed to manipulate. They do not employ reporters. They employ actors and they stage events for public display. Let's talk about this serious situation and how to avoid manipulation.



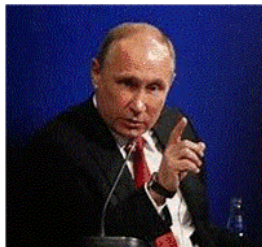
What it was like Growing Up in 1960's and 1970's America

Here is my narrative of some stories of what it was like growing up as a kid in the 1960's and the 1970's. It was a time of freedom, and we took full advantage of it. I'll tell you what. I discuss a little about the clothing, friendships, and relationships of that time.



What High School Taught me About Democracy

America was formed as a Republic. Yet, for many decades we were brought up in the notion that the United States is the greatest "Democracy" in the history of the world. I have my own opinions about that. Democracy is "mob rule". Let's talk about it from what I learned in my High School.



The New Global Order and How to Survive It

The world is polarized. There are two distinctly different ideologies. One is the traditional conservative ideology. The other is the progressive globalist oligarchy that controls just about everything in the West. The future of the world hangs in the resolution of these two ideologies. We discuss that here.



How to take control of your life by using the Rule of Eight.

Here we discuss the "rule of eight" as a method for controlling spending and budgeting for a household. It is a simple rule, and works 100% of the time. Here we discuss it in the three parts or aspects of use, and apply it to purchases by quantity, quality and frequency.

Stories that Inspired Me

Here are reprints in full text of stories that inspired me, but that are nearly impossible to find in China. I place them here as sort of a personal library that I can use for inspiration. The reader is welcome to come and enjoy a read or two as well.



Robert Heinlein's "The Long Watch" – Full Text

What if you were in charge of all the nuclear weapons in the world, and your base Commander checks your political leanings to see if you are willing to conduct a coup d'état. What would you do? Would you go along with the opportunity, or would you be a lone hero?



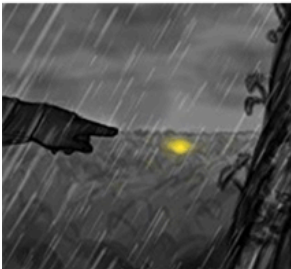
Kaleidoscope (Full Text) A Story by Ray Bradbury

This is the full text of a short story by Ray Bradbury. It concerns the final moments of a crew of spacemen as they drift about in space helplessly after their space rocket explodes. They know that no one can help them as they await their final moments tumbling through space...



Here There Be Tygers by Ray Bradbury

When I was younger I used to read science fiction stories for pleasure. They filled my minds with adventure and passion. One of my favorite authors was Ray Bradbury and here is a complete reprint of one of the stories that has influenced me to be the man that I am today. Enjoy.



The Long Rain (Full Text) by Ray Bradbury

This is the full text of a most excellent science fiction story. It concerns a group of survivors from a crashed spaceship stranded on a Hellish rainy planet; the tropical cloud-covered Venus. They survive the crash, but have to battle the forces of nature and their own passions.

 [Link](#)

 [Link](#)

 [Link](#)

 [Link](#)



The April Witch (Full Text) by Ray Bradbury

This is a story written by Ray Bradbury that describes what might happen when a young teenaged girl has the power to enter the body of another person. What if, forbidden to experience love and relationships with “normal” people, she uses magic to experience forbidden pleasures...



The Last Night of the World (Full Text) by Ray Bradbury

What would you do if you knew that today would be your last day alive? How would you spend it? Would you go running in the streets hysterically? Would you spend it quietly with loved ones? Would you play games on the computer, or watch television? Well, here is a story that explores this interesting situation.



The Flying Machine (Full Text) by Ray Bradbury

One of the most common questions that people ask me is why won't the government tell the people about the existence of extraterrestrials. Well, this story is the best reason that I can think of. This is a story by Ray Bradbury and it is called "The Flying Machine".



October 2026: The Million-Year Picnic (Full Text) by Ray Bradbury

Here is the full story of a man and his family that manages to escape from the madness on Earth. They escape and make their way to Mars to start anew. This is their story, and maybe ours in the near future.



All Summer in a Day (Full Text) by Ray Bradbury

This story takes place in a school room on Venus. Not the Venus that we know, but a wet and rainy Venus. It is a world with constant rain with downpours that lasts for years. However, once every seven years, the sun comes out, and when it comes out the children can see what it is like... or not...



The Smile (Full Text) by Ray Bradbury

This story takes place where the United States lies in ruin. The people behave as animals; rats that scurry in/out of the ruined buildings and avoid the radiation and biological contamination. The survivors of Armageddon truly resent their past; the people who laid their world to waste. Only a single smile remains.

 [The menace from Earth](#)

 [Delilah and the Space Rigger](#)

 [Life-Line](#)

 [The Tax-payer](#)

 [The Pedestrian](#)

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